

Skills Summary

Related Rates and Optimization: Diagnosing the Setup

Use this reference while completing your worksheet or when studying.

Skill 1 — Differentiate the Relation

Order of operations for a setup: relate $\rightarrow$ eliminate $\rightarrow$ differentiate $\rightarrow$ substitute.

- Use the geometry to remove every variable but one *before* differentiating.
- A quantity with a non-zero rate must stay a symbol; only true constants may be written as numbers.
- Similar triangles are the usual eliminator for cones: r/h is fixed by the tank's shape.

Example: A cylinder always has $h = 2r$. Write V in terms of r and differentiate.

Step 1. Two variables cannot both survive into one equation with a single given rate, so eliminate h using the fixed shape ratio first.

Step 2. $V = \pi r^2 h = \pi r^2(2r) = 2\pi r^3$, giving $\frac{dV}{dr} = 6\pi r^2$.

Try it: A cone always has $h = 4r$, so $V = \frac{4}{3}\pi r^3$. Find dV/dr .

check: $4\pi r^2$

Skill 2 — Set Up a Related-Rates Equation

Chain rule on every varying symbol: $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$.

Sphere: $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. Pythagoras: $x^2 + y^2 = c^2 \Rightarrow \frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt}$.

Solve for the wanted rate symbolically, then substitute — that single habit prevents the inverted ratio.

Example: A sphere's volume grows at 50 per second. Find dr/dt when $r = 4$.

Step 1. The given rate is volume and the wanted rate is radius, so differentiate the volume relation and solve for dr/dt before touching a number.

Step 2. $\frac{dr}{dt} = \frac{dV/dt}{4\pi r^2} = \frac{50}{4\pi(4)^2} = 0.249$.

Try it: A 13-ft ladder's base slides out at 3 ft/s. Find dy/dt when the base is 5 ft out.

check: -1.25

Skill 3 — Set Up an Optimization Problem

Write the constraint from the picture, not from memory.

- Objective = the quantity being maximised or minimised; constraint = the resource that is fixed.
- A pen against a wall fences three sides: $x + 2y = P$, not $2x + 2y = P$.
- Use the constraint to make the objective a function of one variable, then set the derivative to zero.
- State the physical domain with the setup; it is what lets you reject a critical value later.

Example: A closed rectangle has perimeter 40. Maximise its area.

Step 1. Perimeter is the fixed resource and area is the objective, so use the constraint to write area with one variable.

Step 2. $2x + 2y = 40 \Rightarrow y = 20 - x$, so $A = x(20 - x)$ and $A' = 20 - 2x = 0$ gives $x = 10$.

Try it: 60 ft of fence, three sides against a wall ($x + 2y = 60$). Which x maximises the area?

check: $08 = x$